

Task -14

14-7-26

Topic : Jenkins Installation and Configuration on AWS EC2 and

Step 1: Launch an EC2 Instance

- Log in to the AWS Management Console.
- Go to EC2 → Launch Instance.
- Enter the instance name.
- Choose t2.micro (Free Tier eligible).
- Select or create a Key Pair (.pem)
- Launch the instance.

✔ **Success**
Successfully initiated launch of instance (i-0908a0caaaef4b55e)

Step 2: Connect to the EC2 Instance

```
PS C:\Users\Karthik Reddy\downloads> ssh -i "rrr.pem" ubuntu@ec2-3-101-155-14.us-west-1.compute.amazonaws.com
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1006-aws x86_64)

* Documentation:  https://docs.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:       https://ubuntu.com/pro
```

Step 3: Update the Server

Command : **sudo apt update**

Step 8: Install Jenkins

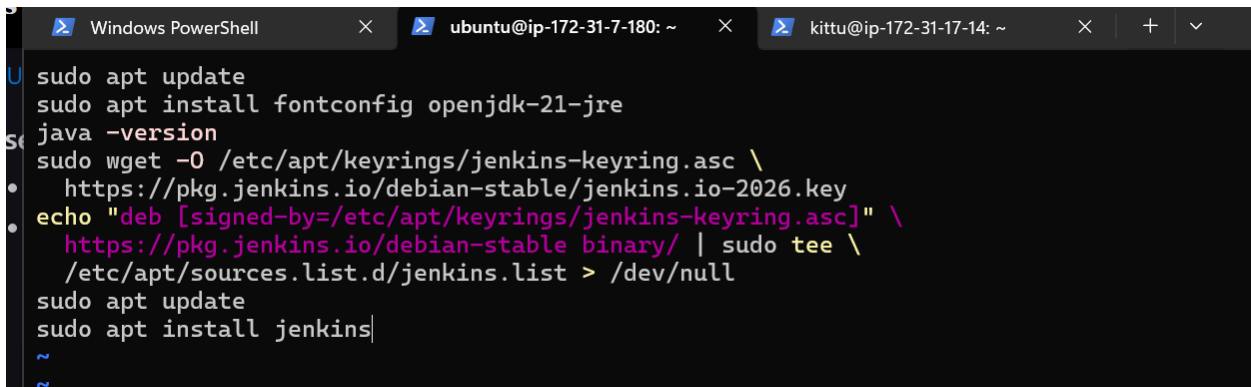
Create a file by using command: **sudo adduser kittu**

```
ubuntu@ip-172-31-7-180:~$ touch kkk.sh
ubuntu@ip-172-31-7-180:~$ ls
kkk.sh
```

Open the file by using the command: **sudo vi kkk.sh**

```
ubuntu@ip-172-31-7-180:~$ sudo vi kkk.sh
"kkk.sh" 10l 400B written
```

Take the command from the installation Linux documentation Copy the code paste it



```
U sudo apt update
S sudo apt install fontconfig openjdk-21-jre
java -version
sudo wget -O /etc/apt/keyrings/jenkins-keyring.asc \
https://pkg.jenkins.io/debian-stable/jenkins.io-2026.key
echo "deb [signed-by=/etc/apt/keyrings/jenkins-keyring.asc] \
https://pkg.jenkins.io/debian-stable binary/" | sudo tee \
/etc/apt/sources.list.d/jenkins.list > /dev/null
sudo apt update
sudo apt install jenkins|
~
~
```

Give the permissions to the file by using the command:

sudo chmod 777 kkk.sh

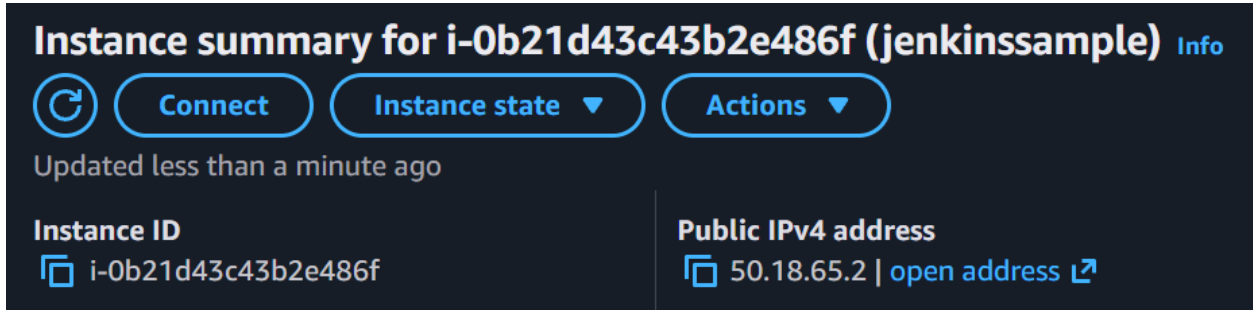
Go check the installation of Linux choose the command:

bash kkk.sh

```
ubuntu@ip-172-31-7-180:~$ bash kkk.sh
Hit:1 http://us-west-1.ec2.archive.ubuntu.com/ubuntu resolute InRelease
Get:2 http://us-west-1.ec2.archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:3 http://us-west-1.ec2.archive.ubuntu.com/ubuntu resolute-backports InRelease [136 kB]
Get:4 http://us-west-1.ec2.archive.ubuntu.com/ubuntu resolute/universe amd64v3 Packages [16.3 MB]
Get:5 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Get:6 http://us-west-1.ec2.archive.ubuntu.com/ubuntu resolute/universe Translation-en [6329 kB]
Get:7 http://us-west-1.ec2.archive.ubuntu.com/ubuntu resolute/universe amd64 Components [4556 kB]
Get:8 http://security.ubuntu.com/ubuntu resolute-security/main amd64v3 Packages [277 kB]
Get:9 http://us-west-1.ec2.archive.ubuntu.com/ubuntu resolute/universe amd64 c-n-f Metadata [313 kB]
Get:10 http://us-west-1.ec2.archive.ubuntu.com/ubuntu resolute/multiverse amd64v3 Packages [291 kB]
Get:11 http://us-west-1.ec2.archive.ubuntu.com/ubuntu resolute/multiverse Translation-en [127 kB]
Get:12 http://us-west-1.ec2.archive.ubuntu.com/ubuntu resolute/multiverse amd64 Components [50.0 kB]
```

Jenkins Sudo Permission: Before and After Configuration

Open AC2 click on Instance Copy the public ID:



Instance summary for **i-0b21d43c43b2e486f (jenkinssample)** [Info](#)

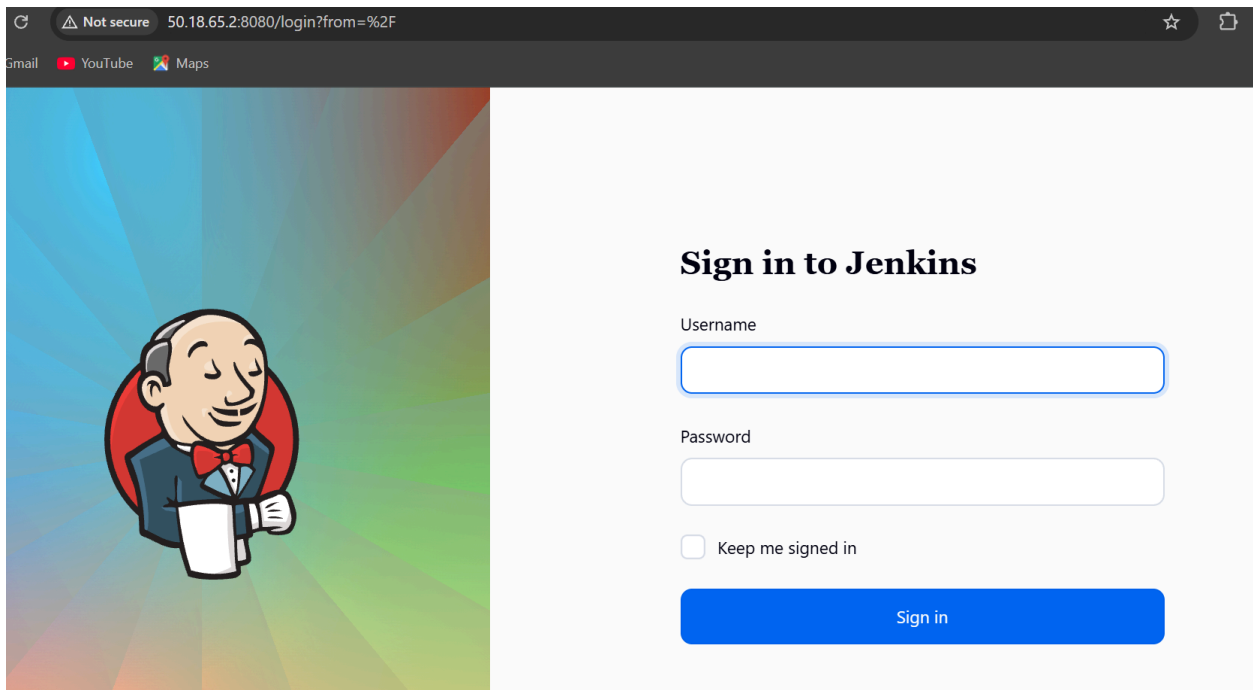
[Refresh](#) [Connect](#) [Instance state ▼](#) [Actions ▼](#)

Updated less than a minute ago

Instance ID  i-0b21d43c43b2e486f	Public IPv4 address  50.18.65.2 open address 
---	---

Paste it on Google:

50.18.65.2:8080 After enter the page will open like this



Browser address bar: [50.18.65.2:8080/login?from=%2F](#)

Sign in to Jenkins

Username

Password

☐ Keep me signed in

[Sign in](#)

Enter the user name and password and Sign in to the account

Enter the item name like Job and the item type after press OK

New Item

Enter an item name

Select an item type

 Pipeline
Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.

 Freestyle project
Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.

 Multi-configuration project
Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.

 Folder

OK

Tab Build steps and add Build steps and execute cell

Build Steps

Automate your build process with ordered tasks like code compilation

+ Add build step

 Filter

Execute Windows batch command

Execute shell

Write the command like

Execute shell ?

Command

See [the list of available environment variables](#)

```
whoami
```

```
echo karthik
```

```
sudo apt update
```

Started by user [karthik Kumar Reddy A](#)

Running as SYSTEM

Building in workspace /var/lib/jenkins/workspace/frist job

[frist job] \$ /bin/sh -xe /tmp/jenkins13393148364868753500.sh

+ whoami

jenkins

+ echo karthik

karthik

+ sudo apt update

sudo: I'm sorry jenkins. I'm afraid I can't do that

Build step 'Execute shell' marked build as failure

Finished: FAILURE

Open the terminal Log in in server:

```
PS C:\Users\Karthik Reddy\downloads> ssh -i "rtr.pem" ubuntu@ec2-3-101-155-14.us-west-1.compute.amazonaws.com
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1006-aws x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro
```

Create the user by using the command:

sudo adduser kittu

```
ubuntu@ip-172-31-17-14:~$ sudo adduser kittu
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for kittu
Enter the new value, or press ENTER for the default
  Full Name []:
  Room Number []:
  Work Phone []:
  Home Phone []:
  Other []:
Is the information correct? [Y/n]
```

Enter the commands to give the permissions

Commands : **cd /etc/ssh**

ls

```
ubuntu@ip-172-31-17-14:~$ cd /etc/ssh
ubuntu@ip-172-31-17-14:/etc/ssh$ ls
```

Command : **ls** to see the files

```
ubuntu@ip-172-31-17-14:/etc/ssh$ ls
moduli      ssh_config.d  ssh_host_ecdsa_key.pub  ssh_host_ed25519_key.pub  ssh_host_rsa_key.pub  sshd_config
ssh_config  ssh_host_ecdsa_key  ssh_host_ed25519_key  ssh_host_rsa_key  ssh_import_id  sshd_config.d
```

To give the permission

Command : **sudo vi sshd_config**

```
ubuntu@ip-172-31-17-14:/etc/ssh$ sudo vi sshd_config
ubuntu@ip-172-31-17-14:/etc/ssh$ sudo vi sshd_config
```

```
#LoginGraceTime 2m
#PermitRootLogin prohibit-password
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10

PubkeyAuthentication yes

# Expect .ssh/authorized_keys2 to be disregarded by default in future.
#AuthorizedKeysFile .ssh/authorized_keys .ssh/authorized_keys2

#AuthorizedPrincipalsFile none

#AuthorizedKeysCommand none
#AuthorizedKeysCommandUser nobody
|
# For this to work you will also need host keys in /etc/ssh/ssh_known_hosts
#HostbasedAuthentication no
# Change to yes if you don't trust ~/.ssh/known_hosts for
# HostbasedAuthentication
#IgnoreUserKnownHosts no
# Don't read the user's ~/.rhosts and ~/.shosts files
#IgnoreRhosts yes
```

To change the permissions enter the command:

cd sshd_config.d

ls

- After the Enter the ls command it will show like
“60-cloudimg-settings.conf” copy it
- Then Enter the command : **sudo vi “paste the copied one”**
After that it will ask permission like this first it have the permission like
“NO” want to change it as “YES”

```
ubuntu@ip-172-31-17-14: ~$  
PasswordAuthentication| yes  
~  
~  
~  
~  
~  
~
```

After the completing want restart it

Command : **"sudo systemctl restart ssh"**

Update the by using the command:

sudo apt update

```
Last login: Tue Jul 14 05:37:31 2020 from 152.57.100.205  
ubuntu@ip-172-31-17-14:~$ sudo apt update  
Hit:1 http://us-west-1.ec2.archive.ubuntu.com/ubuntu resolute InRelease  
Get:2 http://us-west-1.ec2.archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]  
Hit:3 http://us-west-1.ec2.archive.ubuntu.com/ubuntu resolute-backports InRelease  
Get:4 http://us-west-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64v3 Packages [335 kB]  
Get:5 http://us-west-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main Translation-en [87.9 kB]  
Get:6 http://us-west-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [53.1 kB]  
Get:7 http://us-west-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/universe amd64v3 Packages [204 kB]  
Get:8 http://us-west-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/universe Translation-en [63.2 kB]  
Ign:9 https://pkg.jenkins.io/debian-stable binary/ InRelease  
Get:10 http://us-west-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Components [153 kB]  
Hit:11 https://pkg.jenkins.io/debian-stable binary/ Release  
Get:13 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]  
Get:14 http://security.ubuntu.com/ubuntu resolute-security/main amd64v3 Packages [277 kB]  
Get:15 http://security.ubuntu.com/ubuntu resolute-security/main Translation-en [72.7 kB]  
Get:16 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [31.2 kB]  
Get:17 http://security.ubuntu.com/ubuntu resolute-security/universe amd64v3 Packages [133 kB]
```


Enter the command : **sudo visudo**

```
ubuntu@ip-172-31-17-14:~$ sudo visudo
```

Change the permission like this

```
# This file MUST be edited with the 'visudo' command as root
#
# Please consider adding local content in /etc/sudoers.d/ in
# directly modifying this file.
#
# See the man page for details on how to write a sudoers file
#
Defaults      env_reset
Defaults      mail_badpass
Defaults      secure_path="/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin"

# This fixes CVE-2005-4890 and possibly breaks some versions
# (#1011624, https://bugs.kde.org/show_bug.cgi?id=452532)
Defaults      use_pty
```

```
# Allow members of group sudo to execute any command
%sudo  ALL=(ALL:ALL) ALL
kittu  ALL=(ALL) NOPASSWD:ALL
jenkins ALL=(ALL) NOPASSWD:ALL
```

After that **ctl +x** save it and exit “y”

Open new tab on terminal:

Connect to the server and copy the ip address paste it

Command : **ssh kittu@50.18.65.2**

```
PS C:\Users\Karthik Reddy> ssh kittu@50.18.65.2
kittu@50.18.65.2: Permission denied (publickey).
PS C:\Users\Karthik Reddy> ssh kittu@50.18.65.2
kittu@50.18.65.2's password:
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1006-aws x86_64)

* Documentation:  https://docs.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:       https://ubuntu.com/pro
```

```
kittu@ip-172-31-17-14:~$ sudo visudo
sudo: I'm sorry kittu. I'm afraid I can't do that
kittu@ip-172-31-17-14:~$ |
```

After that go to the website and reload the page: **“previous built”**

Console

```
Started by user karthik Kumar Reddy A
Running as SYSTEM
Building in workspace /var/lib/jenkins/workspace/frist job
[frist job] $ /bin/sh -xe /tmp/jenkins13393148364868753500.sh
+ whoami
jenkins
+ echo karthik
karthik
+ sudo apt update
sudo: I'm sorry jenkins. I'm afraid I can't do that
Build step 'Execute shell' marked build as failure
Finished: FAILURE
```

“Next build”

Console

[Download](#)[Copy](#)

```
Started by user karthik Kumar Reddy A
Running as SYSTEM
Building in workspace /var/lib/jenkins/workspace/frist job
[frist job] $ /bin/sh -xe /tmp/jenkins2318672703478823534.sh
+ whoami
jenkins
+ echo karthik
karthik
+ sudo apt update

WARNING: apt does not have a stable CLI interface. Use with caution in scripts.

Hit:1 http://us-west-1.ec2.archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 http://us-west-1.ec2.archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:3 http://us-west-1.ec2.archive.ubuntu.com/ubuntu resolute-backports InRelease
Ign:4 https://pkg.jenkins.io/debian-stable binary/ InRelease
Hit:5 https://pkg.jenkins.io/debian-stable binary/ Release
Hit:7 http://security.ubuntu.com/ubuntu resolute-security InRelease
Reading package lists...
```

